Governed Dataset Onboarding and Lifecycle Routing for AI Training Systems

Dataset Passport and State-Based Lifecycle Framework

Compatible with Access-PoD Governance and Q-Pathformer Execution Environments

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Author: Anthony Siljeg
Affiliation: Alnotrea Labs / Policy of Developments - Access PoD *Initiative*.

Document Scope
This technical report defines a governance framework for dataset onboarding, lifecycle routing, and dataset passport metadata used in AI training environments.
The report establishes governance semantics and lifecycle principles but does not prescribe a specific machine learning architecture, training pipeline, or execution environment.
Execution systems such as Q-Pathformer or other training architectures may consume lifecycle-governed datasets produced under this framework, but such systems do not define governance authority within the Access-PoD model.

This document extends the governance framework established in APOD-TR-002 (Version A), the architecture illustration described in APOD-TR-003 (Version B), and the onboarding discipline defined in APOD-TR-004 (Version C). It introduces a dataset lifecycle and passport model designed to enable governed transformation of inspection artifacts and other structured records into machine-readable datasets for AI training environments.

TR-002 → TR-003 → TR-004 → TR-007
Concept → Architecture → Onboarding → Dataset Lifecycle

Abstract

Artificial intelligence systems require large volumes of training data, yet the sources of that data increasingly originate from governance artifacts, inspection records, simulation outputs, and operational knowledge rather than conventional datasets. Converting these materials into machine-readable training inputs introduces a governance challenge: provenance, authorization scope, and lifecycle constraints must be preserved while enabling controlled use in model development.

This paper introduces a governed dataset onboarding framework designed to bridge that gap. The framework defines how inspection-grade artifacts and other governed data sources can be transformed into structured datasets while maintaining lineage, authorization boundaries, and containment controls. Central to the approach is the **Dataset Passport**, a machine-readable declaration that records provenance, authorized use, and routing constraints for each dataset unit.

The model also introduces a **multi-state dataset lifecycle** in which datasets transition through controlled stages of evaluation, runtime use, and durable training eligibility. This lifecycle prevents uncontrolled dataset mutation and ensures that promotion into persistent training corpora occurs only through explicit governance review.

The framework operates under the governance principles defined in Access-PoD Version B and Version C. It does not confer dataset certification, compliance approval, or trust designation. Instead, it provides a structured mechanism for traceability, containment, and controlled dataset evolution. While compatible with a variety of AI training environments, the framework may optionally interoperate with execution systems such as Q-Pathformer or other machine learning architectures capable of consuming lifecycle-governed datasets.

By separating governance authority from training execution, the model enables organizations to transform governed artifacts into usable AI datasets without sacrificing transparency, auditability, or operational control.

Contents

Abstract 1

1. Introduction 6

2. Relationship to Existing Papers 7

Relationship to Version A 7

Relationship to Version B 7

Relationship to Version C 7

3. Dataset Governance Problem 8

4. Dataset Lifecycle Model 9

Figure 1 — Governed Dataset Lifecycle Model 10

4.1 State 3 — Runtime and Synthetic Dataset Layer 11

4.2 State 2 — Transition and Validation Layer 11

4.3 State 1 — Durable Training Dataset Layer 12

4.4 Promotion Between States 13

4.5 Lifecycle Benefits 13

4.6 Teacher–Student Signal Model 14

5. Dataset Passport Model 15

5.1 Purpose of the Dataset Passport 15

5.2 Passport Structure 16

Figure 2 — Dataset Passport Structure 16

5.3 Provenance Anchors 17

5.4 Authorization Scope 18

5.5 Lifecycle State Declaration 18

5.6 Lineage References 19

5.7 Optional Execution Metadata 19

5.8 Example Dataset Passport 20

5.A - Dataset Containment Principle 21

6. Dataset Authorization and Routing 22

6.1 Authorization Scope 22

6.2 Default Routing Behavior 23
6.3 Routing Conditions 23
6.4 State 2 Signal Convergence 24
6.5 Governance Control of Routing 25
7. Dataset Lineage and Traceability 25
7.1 Purpose of Dataset Lineage 26
7.2 Informative Nature of Lineage Records 26
7.3 Lineage Anchors 27
7.4 Dataset Versioning 27
7.5 Policy–Dataset Lineage 27
7.6 Lineage within the Dataset Lifecycle 28
7.7 Lineage and Governance Oversight 28
8. Controlled Dataset Promotion 29
8.1 Promotion Principles 29
8.2 Promotion from State 3 to State 1 30
8.3 Dataset Conversion During Promotion 30
8.4 Authorization Filtering 31
8.5 Promotion Event Recording 31
8.6 Preservation of Historical Versions 31
9. Optional Execution Environments 32
9.1 Execution-Agnostic Governance Model 32
9.2 Q-Pathformer as a Compatible Reference Environment 33
9.3 Optional UGCCMT and Capsule Metadata 34
9.4 Execution Profiles in the Dataset Passport 35
9.5 Separation of Governance and Execution Authority 35
9.6 Controlled Interoperability 36
10. Governance Boundaries and Non-Claims 36
10.1 No Dataset Certification 37
10.2 No Trust or Assurance Claims 37

10.3 No Compliance Determination 37
10.4 No Transfer of Governance Authority 38
10.5 No Automatic Dataset Promotion 38
10.6 No Implied Model Safety 38
10.7 Preservation of Governance Independence 38
11. Implementation Guidance 39
11.1 Artifact Ingestion 39
11.2 Dataset Conversion 40
11.3 Dataset Passport Generation 40
11.4 Lifecycle Routing 40
11.5 Authorization Filtering 41
11.6 Dataset Promotion Workflow 41
11.7 Optional Execution Integration 42
11.8 Repository Tooling 42
12. Example Dataset Passport 42
12.1 Example Dataset Passport Record 43
12.2 Signal Extraction Example 45
12.3 Lifecycle Interpretation 45
12.4 Optional Execution Metadata 46
12.5 Traceability Outcome 46
13. Future Work 46
13.1 Dataset Converter and Authorization Tooling 47
13.2 Reference Implementations for Execution Environments 47
13.3 Three-State Dataset Learning Model 48
13.4 Client and Sector-Specific Exemplars 48
13.5 Context Window Optimization Across Dataset States 49
13.6 Continuous Dataset Evolution 49
Closing Perspective 50
References 51

1. Introduction

Artificial intelligence development depends on the continuous availability of high-quality datasets. Traditionally, such datasets are constructed through explicit collection and curation processes designed specifically for model training. However, an increasing proportion of potentially valuable training material originates outside traditional dataset pipelines. Governance artifacts, inspection records, simulation outputs, operational logs, and institutional knowledge all contain structured signals that may contribute to AI system development.

Transforming these materials into training datasets introduces several governance challenges. Without explicit controls, provenance information may be lost, authorization boundaries may become unclear, and datasets may be reused in ways that exceed the intent of their original context. These risks are particularly significant in environments where datasets originate from regulatory analysis, organizational decision processes, or policy evaluation activities.

Access-PoD establishes a governance framework for AI systems based on evidence-driven accountability, traceable artifacts, and policy-bound execution environments. Version B introduces an architecture-illustration showing how governance controls may interact with optional AI execution substrates, while Version C defines an onboarding discipline that emphasizes containment, provisional trust, and revocation readiness for emerging AI deployments.

This paper extends those foundations by addressing a practical gap between governance artifacts and AI training pipelines. It proposes a structured method for converting governed artifacts into machine-readable datasets while preserving provenance and authorization constraints.

The proposed approach introduces two key concepts.

The first concept is the **Dataset Passport**, a structured record describing the provenance, authorization scope, and governance status of a dataset unit. The passport acts as a declaration of dataset identity and permitted usage without implying certification, trust designation, or operational readiness.

The second concept is a **multi-state dataset lifecycle**, which separates provisional runtime datasets from durable training corpora. Through controlled routing and promotion rules, datasets can evolve from exploratory or contextual use into stable training assets while maintaining lineage continuity.

Together, these mechanisms provide a governance-aligned pathway for transforming inspection artifacts and other governed materials into usable AI datasets.

2. Relationship to Existing Papers

The framework described in this document is designed to operate within the governance architecture established by the Access-PoD research series. It does not introduce a new governance authority and does not alter the normative definitions provided in earlier publications. Instead, it provides an operational extension focused on dataset lifecycle management.

Relationship to Version A

Access-PoD Version A defines the foundational governance model for AI systems, establishing terminology, assurance semantics, and the principle that trust in AI systems must be grounded in verifiable evidence rather than declarative claims. Version A provides the conceptual framework within which governance artifacts are generated and evaluated.

The dataset onboarding framework described in this paper assumes that artifacts entering the pipeline originate from processes operating under those principles. Consequently, dataset passports record provenance and lineage but do not imply any change to the assurance semantics defined in Version A.

Relationship to Version B

Version B introduces an architecture-specific illustration demonstrating how governance processes may interact with AI execution environments. In that illustration, Access-PoD acts as a policy-driven orchestration layer that governs the interaction between data assets, computational resources, and model training activities.

The dataset onboarding framework extends that illustration by introducing a controlled pathway between governance artifacts and training datasets. In this context, the onboarding process functions as a preparatory stage that organizes artifacts into structured datasets while maintaining traceability and policy alignment.

Execution environments referenced in Version B, including systems such as Q-Pathformer, remain optional substrates for model training. The dataset onboarding model described here is designed to remain independent of any specific training architecture.

Relationship to Version C

Version C introduces a governance discipline for onboarding AI systems under uncertainty. The onboarding model emphasizes containment, provisional trust, telemetry continuity, and revocation readiness. Systems entering operational environments must do so under controlled

conditions that allow governance authorities to observe behavior and respond to emerging risks.

The dataset lifecycle framework described in this paper complements that approach by applying similar principles to datasets themselves. Datasets entering AI training environments may also originate under uncertain conditions, particularly when derived from inspection artifacts or operational records.

For this reason, the onboarding process routes datasets into controlled lifecycle states that allow evaluation and containment before they become durable training assets. Promotion from provisional states into persistent training datasets requires explicit governance review and version issuance.

Relationship to the Q-Pathformer Multi-State Training Concept

The three-state dataset lifecycle model described in this document is conceptually aligned with earlier work describing multi-state machine learning architectures.

The original architectural concept was introduced in December 2024 in the publication *Q-Pathformer: Multi-state Machine Learning and LLM Training*, which describes a training architecture operating across three interacting states for pre-training, vector convergence, and fine-tuning processes.

Reference document: QPATH-ARCH-001
(Q-Pathformer: Multi-state Machine Learning Architecture)

In that architecture, data flows dynamically across three processing states that organize exploratory signals, trained vector relationships, and finalized training outputs.

The lifecycle model defined in this report applies a similar three-state separation to dataset governance rather than model execution. The purpose is to preserve containment and traceability as datasets evolve from provisional runtime signals to durable training corpora.

Within the Access-PoD research series, governance authority remains defined by the Access-PoD framework publications (Version A through Version C and subsequent reports). Execution architectures such as Q-Pathformer are treated as optional implementation environments that may consume lifecycle-governed datasets but do not determine governance authority.

---

3. Dataset Governance Problem

The rapid growth of AI development has expanded the range of materials that may contribute to training datasets. Beyond traditional curated corpora, many organizations now possess

extensive collections of documents, analytical records, and simulation outputs that contain structured insights relevant to machine learning.

Examples include policy analysis reports, governance inspection records, engineering design reviews, operational simulations, and compliance documentation. While these artifacts often contain valuable information, they are rarely structured in a format suitable for direct use in AI training pipelines.

Attempting to convert such artifacts directly into datasets can introduce several governance risks.

First, the provenance of individual dataset elements may become unclear. Without explicit lineage tracking, it becomes difficult to determine the origin of training signals or to verify whether the data was collected under appropriate authorization conditions.

Second, the intended use of the data may be exceeded. Artifacts created for analytical or governance purposes may contain context-specific reasoning that should not automatically become part of a durable model training corpus.

Third, datasets derived from evolving operational environments may change over time. Without versioned lifecycle management, training datasets may be overwritten or mutated in ways that obscure the historical record of how models were developed.

These challenges highlight the need for a structured mechanism that preserves provenance and authorization constraints while enabling controlled dataset transformation.

The framework proposed in this paper addresses these challenges through three mechanisms:

1. **Dataset Passports**, which declare provenance and authorization scope for each dataset unit.
2. **State-based dataset routing**, which separates provisional runtime datasets from durable training datasets.
3. **Controlled promotion rules**, which require governance review before datasets enter persistent training corpora.

Together, these mechanisms create a governance-aligned bridge between inspection artifacts and machine-readable training datasets.

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4. Dataset Lifecycle Model

AI training pipelines typically assume that datasets are static resources prepared specifically for model development. In practice, however, many valuable data sources originate as governance

artifacts, operational records, or analytical outputs that were not originally intended for machine learning. These artifacts may contain valuable signals but also carry contextual limitations, uncertainty indicators, or governance constraints that must be preserved.

To address this challenge, the dataset onboarding framework introduces a **state-based lifecycle model** that separates provisional datasets from durable training corpora. The model ensures that artifacts entering AI training environments can be evaluated and routed according to their provenance, authorization scope, and governance posture.

The three-state lifecycle separation presented here parallels the multi-state machine learning architecture introduced in the Q-Pathformer research work, where training data flows between exploratory pre-training, convergence, and fine-tuning stages.

In this report, the same structural principle is applied to **dataset governance**, enabling controlled progression from provisional runtime datasets to durable training corpora while preserving provenance and authorization boundaries.

The lifecycle is organized into three operational states.

Figure 1 — Governed Dataset Lifecycle Model

The three lifecycle states are governance states describing dataset maturity and authorization status. They are not execution states of a training architecture, although they may conceptually align with multi-state machine learning pipelines.

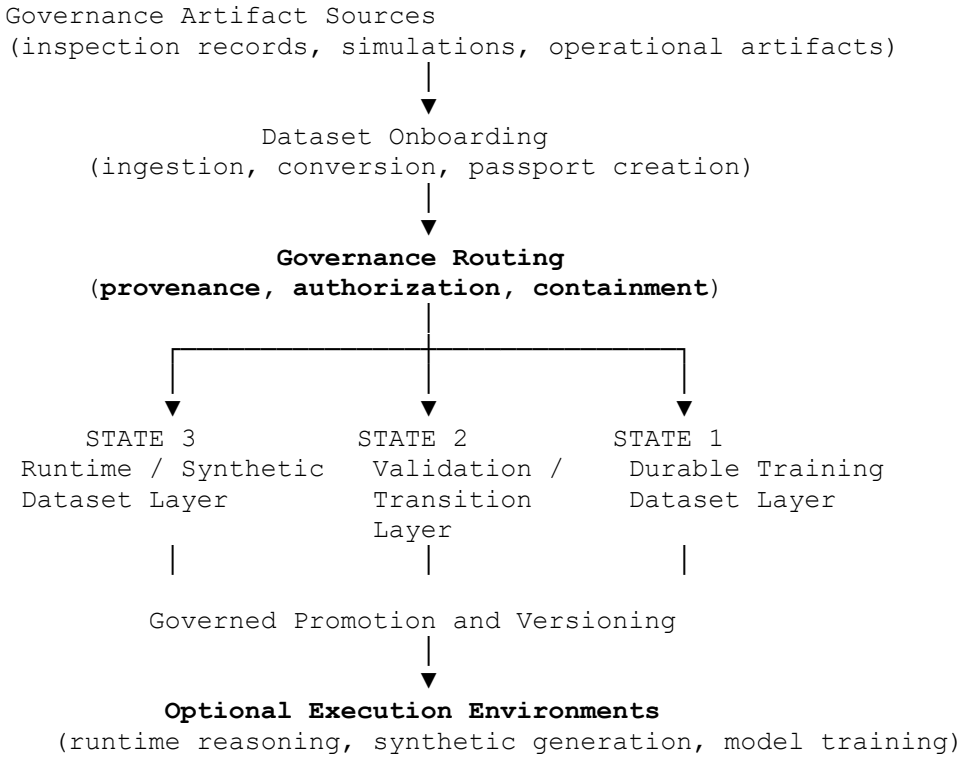


Figure 1 illustrates the dataset lifecycle model introduced in this report. Governance artifacts enter the lifecycle through a dataset onboarding stage in which artifacts are ingested, converted into machine-readable structures, and assigned Dataset Passports. Routing decisions are then applied according to provenance, authorization scope, and containment constraints. Datasets are subsequently managed across three lifecycle states: State 3 for runtime and synthetic dataset use, State 2 for validation and controlled transition, and State 1 for durable training eligibility. Promotion between states remains governed, versioned, and traceable. Optional execution environments may consume lifecycle-routed datasets without acquiring governance authority.

Each state represents a different stage of dataset maturity and authorization.

4.1 State 3 — Runtime and Synthetic Dataset Layer

State 3 is the default entry point for newly onboarded datasets. Datasets routed to this state are considered **provisional or contextual resources** that may be used for runtime reasoning, simulation, or synthetic dataset generation.

State 3 datasets commonly originate from sources such as:

- inspection artifacts produced under governance workflows
- simulation records or scenario analyses
- operational logs and analytical outputs
- newly ingested external datasets awaiting evaluation

Because these datasets may contain contextual reasoning signals rather than validated training knowledge, they are treated as **temporary or exploratory inputs** rather than permanent training material.

Typical uses of State 3 datasets include:

- runtime reasoning support
- synthetic dataset generation
- contextual memory structures
- experimental training signals

State 3 datasets remain traceable through dataset passports and lineage records, ensuring that their origin and authorization scope remain visible even during experimental use.

4.2 State 2 — Transition and Validation Layer

State 2 acts as a **precautionary transition layer** between provisional datasets and durable training datasets.

The purpose of State 2 is to verify that datasets satisfy governance requirements before they become part of a persistent training corpus. This stage introduces structured validation processes that examine dataset provenance, authorization status, and lifecycle constraints.

Typical State 2 validation activities include:

- verifying dataset lineage records
- confirming authorization scope and ownership
- evaluating dataset quality and relevance
- assessing governance constraints and disclosure limits
- ensuring compatibility with organizational policies

Datasets may remain in State 2 for extended periods if governance review or additional validation is required. The transition layer therefore acts as a containment mechanism that prevents premature integration of datasets into model training.

4.3 State 1 — Durable Training Dataset Layer

State 1 represents the highest level of dataset maturity within the lifecycle model. Datasets in this state are considered eligible for inclusion in durable training corpora used for model development or fine-tuning.

To reach State 1, a dataset must pass through governance review and receive explicit authorization for persistent use. Promotion into State 1 is therefore treated as a **controlled lifecycle event** rather than an automatic transformation.

Characteristics of State 1 datasets include:

- stable and versioned dataset structures
- verified provenance and lineage records
- explicit authorization for training use
- compatibility with organizational governance policies
- controlled versioning and update procedures

Importantly, State 1 datasets remain subject to versioned lifecycle management. When datasets are updated or extended, new versions must be issued rather than modifying existing datasets in place. This approach preserves historical traceability and prevents silent mutation of training corpora.

4.4 Promotion Between States

Datasets may transition between lifecycle states through controlled promotion processes. These transitions are designed to maintain governance visibility and ensure that datasets evolve in a traceable and reversible manner.

Typical transitions include:

- **State 3 → State 2**
  when provisional datasets undergo validation or governance review.
- **State 2 → State 1**
  when datasets are formally authorized for durable training use.

Promotion events must create a new dataset version and update the associated dataset passport. Previous versions remain preserved in lineage records to maintain continuity of evidence.

Direct transitions from State 3 to State 1 are discouraged unless governance authorities explicitly approve the dataset and verify that the validation requirements of State 2 have been satisfied.

4.5 Lifecycle Benefits

The multi-state dataset model provides several important benefits for governance-aligned AI development.

First, it prevents provisional artifacts from automatically becoming permanent training datasets. This containment mechanism ensures that datasets derived from inspection records or operational environments can be evaluated before influencing model behavior.

Second, it preserves provenance and authorization boundaries. Dataset passports and lineage records maintain a clear record of how datasets were created, modified, and promoted throughout their lifecycle.

Third, the lifecycle model supports controlled dataset evolution. By requiring explicit promotion events and version issuance, organizations can update training corpora while maintaining historical traceability.

Finally, the lifecycle framework allows AI training pipelines to incorporate diverse data sources without sacrificing governance discipline. Inspection artifacts, simulation outputs, and

operational knowledge can be converted into machine-readable datasets while remaining subject to structured oversight.

4.6 Teacher–Student Signal Model

In addition to lifecycle states, datasets may contain internal signal relationships that influence how information is interpreted during training or runtime reasoning. One useful conceptual model for describing these relationships is the **teacher–student signal structure**.

In this model, datasets are interpreted through two complementary roles.

The **teacher role** represents structured signals derived from dataset artifacts. These signals may include scenario descriptions, analytical observations, governance annotations, or other structured reasoning outputs contained within the dataset.

The **student role** represents the interpretive learning process that extracts patterns from those signals during training or runtime reasoning.

The teacher–student model does not imply a hierarchical authority structure. Instead, it describes how signals contained within a dataset guide the learning process while preserving the contextual limitations of the source artifacts.

Within the dataset lifecycle model, teacher and student signals may appear in different forms depending on the dataset state.

In **State 3**, teacher signals commonly originate from inspection artifacts, simulation outputs, or contextual analytical records. These signals often describe reasoning pathways, uncertainty conditions, and governance constraints rather than finalized conclusions. Student processes interpret these signals to generate contextual understanding, synthetic dataset candidates, or exploratory training signals.

In **State 1**, teacher signals may appear as structured prompts, curated examples, or validated dataset annotations that guide model fine-tuning. Student processes then interpret these signals to produce stable learning patterns suitable for durable training corpora.

The teacher–student model therefore provides a conceptual bridge between artifact-based reasoning and structured training signals. Rather than treating datasets as static repositories of information, the model acknowledges that many datasets function as collections of reasoning signals that guide learning processes.

This perspective is particularly relevant for datasets derived from inspection artifacts or analytical workflows. Such artifacts often contain observations, stress conditions, and contextual notes that are valuable for model development but require careful interpretation.

By recognizing these signals as teacher inputs rather than authoritative conclusions, organizations can transform governance artifacts into training signals while preserving the contextual limits of the original material.

---

5. Dataset Passport Model

The dataset lifecycle model introduced in the previous section requires a structured mechanism for recording dataset provenance, authorization scope, and lifecycle status. To support this requirement, the framework introduces the **Dataset Passport**, a machine-readable record that accompanies each dataset unit entering the onboarding pipeline.

The Dataset Passport acts as a governance-aligned declaration of dataset identity and permitted usage. It records the origin of the dataset, the lifecycle state in which the dataset currently resides, and the authorization conditions under which the dataset may be used.

Importantly, the Dataset Passport does not function as a certification mechanism. It does not grant approval, trust designation, compliance status, or assurance claims. Instead, it provides traceability and operational guidance for dataset routing within the lifecycle model.

Dataset passports are therefore **informative governance artifacts**, not authoritative determinations.

5.1 Purpose of the Dataset Passport

The Dataset Passport fulfills several roles within the dataset onboarding framework.

First, it preserves **dataset provenance**. Each passport records the origin of the dataset artifact and references the canonical records from which the dataset was derived. This ensures that lineage information remains visible even when datasets are transformed into machine-readable structures.

Second, the passport defines **authorization scope**. Organizations may permit datasets to be used for specific purposes such as runtime reasoning, synthetic dataset generation, or model fine-tuning. These permissions are recorded within the passport so that dataset routing mechanisms can enforce the intended usage boundaries.

Third, the passport records **dataset lifecycle status**. The passport identifies whether a dataset currently resides in State 3, State 2, or State 1. This allows training pipelines to treat datasets differently depending on their maturity and authorization level.

Finally, the passport provides a stable identifier for datasets as they evolve. When datasets are updated or promoted between lifecycle states, new passport versions may be issued while preserving links to earlier versions.

5.2 Passport Structure

The structure of the Dataset Passport is illustrated in Figure 2.

Figure 2 — Dataset Passport Structure

The Dataset Passport is a machine-readable governance declaration attached to each dataset entering the lifecycle pipeline.

```
Dataset Passport
│
├── Core Governance Fields
│
│   ├── Dataset Identity
│   │      dataset_passport_id
│   │      dataset_title
│   │      dataset_origin
│   │
│   ├── Provenance Anchors
│   │      source_mode
│   │      canonical_artifacts
│   │      artifact_hashes
│   │      freeze_timestamp
│   │
│   ├── Lifecycle Declaration
│   │      lifecycle_state
│   │      state_transition_history
│   │
│   ├── Authorization Scope
│   │      runtime_reasoning
│   │      synthetic_generation
│   │      fine_tuning
│   │      external_training
│   │
│   └── Lineage References
│          parent_dataset
│          source_sessions
│          policy_references
│
└── Optional Execution Extension
       dataset_clusters
       capsule_identifiers
       training_stage
       execution_environment
```

Figure Interpretation

The Dataset Passport provides the governance metadata required to manage datasets throughout the lifecycle.

The passport records the dataset's identity, provenance anchors, lifecycle state, and authorization scope. Lineage references preserve traceability between dataset versions and the artifacts from which they were derived.

Optional execution metadata may be included when datasets are consumed by training environments such as Q-Pathformer. However, execution metadata does not alter governance authority or lifecycle status.

The passport therefore acts as a **governance boundary**, ensuring that datasets remain traceable and policy-constrained even when used within model training pipelines.

A Dataset Passport contains several categories of information.

These categories are designed to support both governance oversight and automated dataset routing.

Typical fields include:

- dataset identity
- provenance anchors
- lifecycle state
- authorization scope
- lineage references
- optional execution metadata

The passport format is intentionally modular so that organizations can extend it with additional metadata fields without altering the core governance structure.

5.3 Provenance Anchors

Provenance anchors provide cryptographic and documentary references to the artifacts from which the dataset was derived.

In the case of datasets originating from Access-PoD inspection workflows, provenance anchors typically reference **ID_3 frozen artifacts** and their associated hash manifests. These anchors ensure that dataset lineage remains verifiable even after artifacts are transformed into machine-readable structures.

Typical provenance anchors may include:

- source artifact identifiers
- canonical artifact filenames
- cryptographic hashes of frozen artifacts
- freeze timestamps
- governance mode identifiers

Together, these anchors allow auditors or governance authorities to trace a dataset back to its original inspection records.

5.4 Authorization Scope

The authorization section of the Dataset Passport defines the permitted uses of the dataset.

Authorization scopes may vary depending on organizational policies, regulatory requirements, or contractual obligations. The passport records these permissions so that dataset routing systems can enforce them automatically.

Examples of authorization scopes include:

- runtime reasoning use
- synthetic dataset generation
- model fine-tuning
- external model training
- research experimentation

Authorization entries should always remain **revocable and versioned**. If organizational policies change or if dataset conditions evolve, new passport versions can update the authorization scope while preserving the historical record.

5.5 Lifecycle State Declaration

Each Dataset Passport records the lifecycle state in which the dataset currently resides.

The lifecycle state determines how the dataset may be used within AI pipelines.

State declarations include:

- **State 3 — Runtime and synthetic dataset layer**
  used for contextual reasoning and exploratory signals
- **State 2 — Validation and transition layer**
  used for governance review and dataset verification
- **State 1 — Durable training dataset layer**
  authorized for inclusion in persistent training corpora

By recording the lifecycle state in the passport, training systems can automatically route datasets according to their maturity level.

5.6 Lineage References

Datasets often evolve through multiple transformation stages. For example, a dataset derived from inspection artifacts may later be refined into a curated training corpus.

To preserve traceability, the Dataset Passport includes lineage references that link related dataset versions.

Lineage records may reference:

- parent dataset passports
- earlier dataset versions
- policy lineage records
- source artifact identifiers

These references ensure that dataset evolution remains visible and reversible.

Lineage records are **informative documentation**, not indicators of approval or trust status. Their purpose is to support traceability and governance oversight.

5.7 Optional Execution Metadata

The dataset onboarding framework is designed to remain independent of specific AI training architectures. However, execution environments may require additional metadata to organize datasets during training.

For this reason, the Dataset Passport may optionally include execution metadata fields.

Examples include:

- dataset cluster identifiers
- training dataset tags
- dataset classification groups
- execution environment hints

In systems such as Q-Pathformer, these fields may support clustering mechanisms, dataset grouping, or cryptographic provenance encoding. These implementation details remain optional and do not alter the governance semantics of the passport.

5.8 Example Dataset Passport

The following simplified example illustrates a dataset passport derived from an inspection artifact.

```
{
  "dataset_passport_id": "qpds-2026-02-10-001",
  "dataset_title": "Inspection Stress Scenario Signals",
  "source_mode": "Access-PoD ID_3",
  "source_artifact": "module_03_baseline.canonical.json",
  "artifact_hash":
"bd543483467ba8091f05b91ffafb96cc72c5a69c26995c211edcdb6cf3e1637c",
  "freeze_timestamp": "2026-02-28T21:57:07Z",

  "lifecycle_state": "STATE_3",

  "authorization_scope": {
    "runtime_reasoning": true,
    "synthetic_generation": true,
    "fine_tuning": false,
    "external_training": false
  },

  "lineage": {
    "parent_artifact": "ID3_FREEZE_2026-02-10",
    "policy_reference": "APOD-TR-004 Version C"
  }
}
```

This passport indicates that the dataset originates from an inspection artifact and is currently authorized only for runtime reasoning and synthetic dataset generation.

Promotion into durable training datasets would require a new passport version and explicit governance authorization.

5.A - Dataset Containment Principle

The dataset onboarding framework operates under a foundational constraint referred to as the **Dataset Containment Principle**.

This principle ensures that datasets entering AI training environments remain subject to governance constraints throughout their lifecycle and cannot implicitly acquire authority, trust status, or operational approval merely by existing within a dataset pipeline.

Dataset containment applies to all lifecycle states and all dataset transformation processes.

Under this principle, the presence of a dataset within the onboarding framework does not imply:

- dataset approval
- dataset certification
- dataset trustworthiness
- model safety
- compliance determination
- regulatory acceptance

Datasets are treated as **evidence-bearing artifacts**, not authoritative knowledge sources.

Containment ensures that dataset signals derived from governance artifacts, operational records, or analytical outputs remain context-bound and subject to explicit authorization controls before influencing durable model behavior.

Within the lifecycle model, containment is enforced through several mechanisms.

First, datasets enter the lifecycle through a **Dataset Passport**, which records provenance anchors and authorization scope. The passport establishes the governance context in which the dataset may be used.

Second, datasets are routed through lifecycle states that separate provisional datasets from durable training corpora. Newly onboarded datasets typically enter **State 3**, where they may be used for contextual reasoning, simulation, or synthetic dataset generation while remaining under containment.

Third, dataset promotion into **State 1 durable training corpora** requires explicit governance authorization and the issuance of a new dataset version. This process prevents provisional artifacts from automatically influencing long-term model behavior.

Fourth, dataset lineage records maintain traceability between dataset versions and their originating artifacts. This ensures that dataset transformations remain auditable and reversible.

The Dataset Containment Principle therefore provides a governance boundary between **dataset existence and dataset authority**. It ensures that datasets may be explored, analyzed, and transformed while preventing uncontrolled propagation of signals into durable model training.

This containment approach is particularly important when datasets originate from inspection artifacts, simulation outputs, or analytical reasoning records. Such artifacts often contain contextual observations rather than finalized conclusions. Treating them as authoritative training material without containment could introduce unintended bias or misinterpretation into AI systems.

By preserving containment across all lifecycle states, the onboarding framework allows organizations to transform governance artifacts into machine-readable datasets while maintaining operational control and traceable accountability.

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6. Dataset Authorization and Routing

The dataset onboarding framework requires a structured mechanism for determining how datasets are used once they enter the lifecycle pipeline. Dataset authorization and routing define the operational logic that determines whether a dataset may be used for runtime reasoning, synthetic dataset generation, or durable model training.

Authorization decisions are not determined solely by dataset origin. Instead, routing decisions are derived from a combination of factors recorded within the Dataset Passport, including provenance anchors, lifecycle state, authorization scope, and governance constraints.

Dataset routing therefore acts as the operational layer that translates governance declarations into dataset behavior within AI training environments.

6.1 Authorization Scope

The Dataset Passport records the authorization scope assigned to each dataset unit. Authorization scope defines the types of activities for which a dataset may be used.

Typical authorization scopes include:

- runtime reasoning support
- synthetic dataset generation
- model experimentation
- fine-tuning dataset preparation
- external training use

Authorization scopes may differ across datasets depending on the policies of the originating organization or the governance context under which the dataset was created.

For example, datasets derived from inspection artifacts may initially be authorized only for runtime reasoning or exploratory simulation. In such cases, the dataset remains within provisional lifecycle states until governance review determines whether promotion to durable training datasets is appropriate.

Authorization entries are always **revocable and versioned**. If authorization conditions change, the Dataset Passport must be updated and a new dataset version issued.

6.2 Default Routing Behavior

Datasets entering the onboarding pipeline follow default routing rules designed to preserve containment and governance control.

In most cases, newly onboarded datasets are routed to **State 3**, where they may support contextual reasoning, simulation, or synthetic dataset generation while remaining under containment.

Routing to **State 2** occurs when datasets undergo validation or governance review processes. This stage allows organizations to examine dataset provenance, evaluate signal quality, and confirm authorization boundaries.

Routing to **State 1** occurs only after explicit authorization for durable training use. This promotion event typically requires governance review and version issuance.

These routing rules ensure that datasets derived from governance artifacts do not automatically become permanent training corpora.

6.3 Routing Conditions

Routing decisions are typically based on several factors recorded within the Dataset Passport.

These factors may include:

- lifecycle state declaration
- dataset provenance anchors
- authorization scope
- lineage relationships

- governance policy references

Automated routing systems may evaluate these attributes when determining how a dataset should be handled within the lifecycle model.

For example, a dataset passport indicating **State 3 authorization only** would prevent the dataset from entering durable training pipelines. Conversely, a dataset passport authorizing fine-tuning may permit promotion into State 1 after validation.

6.4 State 2 Signal Convergence

Within the dataset lifecycle model, **State 2 functions as the convergence layer for teacher and student signals**.

Teacher signals originate from structured dataset artifacts such as inspection records, scenario analyses, governance annotations, or curated prompts. These signals represent structured reasoning inputs contained within the dataset.

Student signals represent the interpretive learning processes that attempt to extract patterns or insights from those signals during dataset evaluation or training preparation.

In State 2, these signals are examined together to determine whether the dataset demonstrates sufficient coherence and contextual clarity to support further lifecycle progression.

This convergence does not produce authoritative decisions by itself. Instead, it supports the validation processes that inform dataset promotion or continued containment.

Examples of convergence evaluation activities may include:

- assessing whether teacher signals provide consistent reasoning pathways
- examining how student interpretations respond to those signals
- identifying contradictions or uncertainty within the dataset signals
- evaluating whether the dataset supports stable learning patterns

By examining both signal types together, the State 2 layer reinforces the interpretive discipline introduced in Section 4. Rather than treating datasets as static knowledge sources, the system evaluates how reasoning signals interact before allowing them to influence durable model behavior.

6.5 Governance Control of Routing

While routing mechanisms may be implemented through automated systems, governance authority remains external to the routing process itself.

Routing systems interpret dataset passports and apply lifecycle rules, but they do not grant authorization independently. Authorization remains determined by governance policies, organizational review processes, or external regulatory conditions.

For this reason, routing outcomes must remain traceable and reversible. Dataset promotion events, authorization updates, and lifecycle transitions should always produce new dataset versions and corresponding passport updates.

This approach ensures that dataset evolution remains visible within lineage records and that governance oversight can intervene if unexpected behaviors emerge.

---

7. Dataset Lineage and Traceability

The dataset onboarding framework relies on traceability mechanisms that preserve the relationship between datasets, their originating artifacts, and the governance context in which they were created. These relationships are collectively referred to as **dataset lineage**.

Dataset lineage provides a structured method for recording how datasets originate, how they evolve over time, and how they relate to other datasets or governance artifacts. Lineage information allows organizations to reconstruct the historical pathway through which a dataset entered an AI training environment.

Lineage records therefore function as a **documentation layer** within the governance framework. They enable transparency and auditability without implying that a dataset has been approved, certified, or deemed trustworthy.

Within the onboarding framework, lineage operates in conjunction with the Dataset Passport and the lifecycle routing model to ensure that dataset transformations remain visible and reversible.

7.1 Purpose of Dataset Lineage

The primary purpose of dataset lineage is to preserve **traceable continuity** between dataset artifacts and the processes that created or modified them.

AI training datasets often undergo multiple transformations. Raw artifacts may be converted into machine-readable records, refined through validation processes, and eventually incorporated into curated training corpora. Without lineage tracking, it becomes difficult to determine how training signals entered a model or how dataset versions relate to earlier artifacts.

Dataset lineage addresses this challenge by recording:

- the origin of dataset artifacts
- the transformation processes applied to those artifacts
- the relationships between dataset versions
- the governance context associated with each dataset stage

These records allow organizations to reconstruct dataset evolution and verify that dataset transformations occurred under appropriate governance conditions.

7.2 Informative Nature of Lineage Records

Lineage records are **informative documentation**, not indicators of dataset validity or trust status.

Recording the lineage of a dataset does not imply that the dataset has been approved for model training, that the dataset is free from bias or error, or that the dataset meets any regulatory compliance standard.

Instead, lineage records provide contextual information that allows governance authorities and auditors to understand how datasets were created and how they changed over time.

This distinction is important for maintaining the governance boundaries established in earlier publications. Dataset presence within lineage records must not be interpreted as evidence of operational readiness, certification, or assurance.

7.3 Lineage Anchors

Dataset lineage records typically include references to **lineage anchors**. These anchors connect dataset passports to the artifacts or processes that produced them.

Common lineage anchors may include:

- source artifact identifiers
- canonical inspection records
- cryptographic hash manifests
- dataset passport identifiers
- policy references associated with dataset creation

For datasets derived from inspection workflows, lineage anchors may reference **frozen artifacts and their associated hash manifests**. These anchors allow governance authorities to verify that datasets originated from authentic records rather than undocumented transformations.

Lineage anchors therefore serve as structural points of reference within the dataset history.

7.4 Dataset Versioning

Datasets evolve over time as they move through the lifecycle states defined earlier in this document. For example, a dataset initially used for runtime reasoning may later be refined into a curated training dataset.

To preserve traceability, dataset changes should be managed through **versioned updates** rather than in-place modification.

When a dataset changes in ways that affect its content, authorization scope, or lifecycle status, a new dataset passport version should be issued. The new passport references the previous version through lineage records, creating a chain of dataset evolution.

Versioned dataset updates allow organizations to maintain historical visibility into how training corpora were constructed and modified.

7.5 Policy–Dataset Lineage

In addition to dataset lineage, the onboarding framework recognizes the concept of **policy lineage**.

Policy lineage records describe the governance policies or procedural frameworks that influenced dataset creation or transformation. These records help contextualize why certain datasets were produced and under what governance conditions they were authorized.

For example, a dataset may reference the governance framework defined in earlier publications within the Access-PoD research series. Recording such policy relationships helps auditors understand the governance context in which dataset transformations occurred.

As with dataset lineage, policy lineage is informative rather than authoritative. It documents governance context without implying compliance certification or policy enforcement outcomes.

7.6 Lineage within the Dataset Lifecycle

Within the lifecycle model introduced earlier, lineage records play an important role in preserving dataset continuity across lifecycle states.

When datasets transition between states—such as moving from State 3 exploratory datasets to State 1 durable training corpora—lineage records ensure that the transformation remains visible.

The lineage record may include:

- the previous dataset passport identifier
- the transformation event that produced the new dataset
- the governance review process associated with the transition
- the lifecycle state change

These records create a transparent pathway through which dataset evolution can be examined.

7.7 Lineage and Governance Oversight

Lineage mechanisms support governance oversight by providing the evidence needed to evaluate dataset evolution.

Because dataset lineage records capture the relationships between artifacts, datasets, and governance processes, they allow oversight bodies to examine how training datasets were constructed and whether transformations occurred within authorized boundaries.

Lineage records also support corrective action if necessary. If a dataset is later determined to contain problematic signals or to exceed its authorization scope, lineage records allow organizations to identify affected dataset versions and trace their downstream influence.

For this reason, maintaining complete lineage documentation is a key component of governance-aligned dataset management.

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8. Controlled Dataset Promotion

The dataset lifecycle model introduced in earlier sections separates provisional datasets from durable training datasets. To preserve governance control, transitions between lifecycle states must occur through structured **promotion events**.

Dataset promotion refers to the process by which a dataset moves from a provisional or exploratory lifecycle state into a durable training dataset authorized for persistent use.

Promotion events ensure that datasets influencing long-term model behavior are introduced in a controlled and traceable manner.

8.1 Promotion Principles

Dataset promotion operates under several guiding principles.

First, promotion must be **explicit and versioned**. A dataset does not become a durable training dataset simply by existing within a dataset pipeline. Promotion requires the issuance of a new dataset passport version that records the lifecycle transition.

Second, promotion must remain **traceable through lineage records**. The newly promoted dataset must reference its parent dataset and the artifacts from which it was derived.

Third, promotion must preserve **dataset containment**. If portions of a dataset exceed their authorization scope or governance boundaries, those portions must be removed or filtered before the dataset becomes eligible for durable training.

These principles ensure that dataset promotion remains transparent and reversible.

8.2 Promotion from State 3 to State 1

Datasets initially routed to **State 3** are considered provisional resources that may support runtime reasoning, simulation, or synthetic dataset generation. While these datasets may contain valuable signals, they are not automatically eligible for inclusion in durable training corpora.

Promotion from State 3 to **State 1** occurs only after governance validation confirms that the dataset meets the requirements for persistent training use.

Typical promotion steps may include:

1. verifying dataset provenance and lineage anchors
2. evaluating the coherence of teacher and student signals
3. confirming the dataset's authorization scope
4. filtering dataset elements that exceed their permitted usage
5. issuing a new dataset passport reflecting the State 1 lifecycle status

Once these steps are completed, the dataset may be introduced into durable training pipelines.

8.3 Dataset Conversion During Promotion

Promotion may involve a transformation step in which provisional dataset structures are converted into curated training datasets.

During this process, dataset signals collected during earlier lifecycle stages are reorganized into structured training records suitable for model development.

For example, contextual reasoning signals derived from inspection artifacts may be transformed into:

- structured prompts
- scenario-based training examples
- annotated dataset records

These transformations should preserve provenance anchors so that each training signal remains traceable to its original artifact.

The conversion process therefore acts as a bridge between exploratory dataset signals and durable training data.

8.4 Authorization Filtering

Before a dataset becomes eligible for durable training use, it must undergo **authorization filtering**.

Authorization filtering ensures that only dataset elements consistent with governance policies are included in the final dataset.

This filtering process may remove or restrict dataset elements that:

- exceed the dataset's authorization scope
- contain sensitive information not approved for training
- conflict with governance policies or organizational standards

The resulting dataset represents the subset of signals that are authorized for persistent model training.

Authorization filtering therefore acts as the final containment control before a dataset enters State 1.

8.5 Promotion Event Recording

Every promotion event must be recorded within the dataset lineage framework.

Promotion records typically include:

- the parent dataset passport identifier
- the lifecycle state transition
- the dataset version identifier
- the governance policy reference associated with the promotion
- the timestamp of the promotion event

Recording these events ensures that dataset evolution remains transparent and that training datasets can be traced back to their originating artifacts.

8.6 Preservation of Historical Versions

Promotion does not overwrite previous dataset versions.

Instead, new dataset versions are created while earlier versions remain preserved within the lineage record. This approach ensures that the historical pathway of dataset evolution remains visible and that organizations can reconstruct earlier training environments if necessary.

Maintaining historical dataset versions also supports governance oversight by allowing auditors to examine how training datasets changed over time.

---

9. Optional Execution Environments

The dataset onboarding framework described in this document is designed to operate independently of any single model architecture, training substrate, or execution environment. Its primary purpose is to preserve governance context, dataset provenance, lifecycle state, and authorization constraints as datasets move from governed artifacts into machine-readable forms.

For this reason, the framework is **execution-agnostic by design**. A dataset may be onboarded, assigned a Dataset Passport, and routed through lifecycle states without requiring a specific training implementation. This preserves the separation established in earlier Access-PoD publications between governance authority and execution mechanisms. Version B explicitly treats architecture-specific implementations as illustrative and non-authoritative, while Version C preserves containment and controlled onboarding as the governing posture for operational transition.

APOD-TR-004_Vol2_Version-C

At the same time, many organizations will require an execution environment capable of consuming routed datasets, organizing them into training-ready structures, and preserving provenance during training or runtime use. For such cases, the onboarding framework may interoperate with optional execution environments through extension metadata recorded in the Dataset Passport.

9.1 Execution-Agnostic Governance Model

The governance model defined in this document does not depend on a particular machine learning platform. Its core components remain the same regardless of the downstream execution environment:

- Dataset Passports
- lifecycle state declarations
- authorization scope

- lineage references
- promotion controls
- containment constraints

These elements are sufficient to govern dataset onboarding and lifecycle management even when no model training occurs. This design allows the framework to support multiple environments, including internal enterprise pipelines, third-party AI systems, or specialized research architectures.

Execution systems therefore consume governance outputs; they do not replace governance authority. A dataset's presence in a training environment does not alter its authorization scope, lifecycle state, or containment status. Those remain determined by the governance framework and recorded in the Dataset Passport.

9.2 Q-Pathformer as a Compatible Reference Environment

One compatible execution environment for this framework is **Q-Pathformer**, which may serve as an optional reference substrate for governed dataset transformation, clustering, and staged training preparation.

Q-Pathformer is described as a dataset training platform built around reproducibility, audit-grade provenance, modular capsule packaging, and structured dataset curation. Its architecture emphasizes versioned and validated training units, making it compatible with the governance requirements of dataset passports and lifecycle routing.

The Q-Pathformer materials also describe a hybrid architecture in which Access-PoD acts as the policy-driven orchestration layer, while Q-Pathformer provides execution-side mechanisms for dataset grouping, adaptive weighting, and model training. The UGCCMT model further introduces a structured method for partitioning data into clusters, assigning colour-coded categories, encoding metadata into cryptographic salts, and shaping training behavior through weighted measures.

ugccmt v2

Within this document, Q-Pathformer is treated as:

- a compatible execution environment,
- an optional implementation substrate,
- and a possible consumer of Dataset Passport metadata.

It is **not** treated as the source of governance authority.

The conceptual separation between governance and execution is central to the Access-PoD architecture.

Access-PoD publications define governance authority, lifecycle rules, and dataset authorization semantics. Execution environments such as Q-Pathformer implement training pipelines capable of consuming governed datasets and organizing them into training structures.

In this relationship, Access-PoD provides the governance layer while Q-Pathformer provides a potential execution substrate for machine learning workflows. The lifecycle framework defined in this report therefore operates independently of the training architecture while remaining compatible with environments that implement multi-state learning systems.

9.3 Optional UGCCMT and Capsule Metadata

When Q-Pathformer or a similar execution environment is used, the Dataset Passport may include an **optional execution extension**. This extension records implementation-specific metadata needed by the downstream training or runtime system.

In the case of Q-Pathformer, optional metadata may include:

- UGCCMT grouping identifiers
- colour-code classification
- measure profiles
- capsule identifiers
- staged training hints
- synthetic or runtime preparation flags

The UGCCMT model defines six stages—Unit, Group, Colour, Code, Measure, and Train—which together support partitioning, risk classification, metadata encoding, and training-sensitive key derivation. These stages can be represented as optional metadata fields attached to a passport without altering the passport's governance semantics.

Similarly, capsule pipeline metadata may describe how a dataset is organized into reproducible training units for staged execution. Q-Pathformer materials describe capsules as modular training units containing curated data, deterministic labeling, semantic enrichment, and embedded configuration metadata.

These metadata fields are therefore **execution-supporting**, not governance-defining.

9.4 Execution Profiles in the Dataset Passport

To preserve interoperability, the Dataset Passport should distinguish between:

- **core governance fields**, which are always present, and
- **optional execution fields**, which appear only when required by the target environment.

Core governance fields typically include:

- dataset identity
- provenance anchors
- lifecycle state
- authorization scope
- lineage references
- containment and promotion status

Optional execution fields may include:

- execution environment identifier
- clustering or grouping metadata
- training-stage indicators
- capsule pipeline references
- model-specific routing hints

This separation allows the same passport structure to support both governance-only workflows and execution-aware workflows. A dataset may therefore remain fully valid within the onboarding framework even if no execution metadata is present.

9.5 Separation of Governance and Execution Authority

A key principle of this framework is the separation between **governance authority** and **execution behavior**.

Execution environments may consume dataset passports, organize data into training structures, and process optional metadata for clustering or staged learning. However, they do not independently determine:

- whether a dataset is authorized for durable training,
- whether a dataset may be used externally,
- whether a dataset has been promoted into State 1, or
- whether a dataset has achieved any trust, assurance, or compliance status.

These determinations remain external to the execution environment and must be governed through the lifecycle and authorization controls defined earlier in this document.

This separation is particularly important where advanced execution systems introduce cryptographic provenance encoding, synthetic generation, or staged training pipelines. Such mechanisms may improve traceability or training discipline, but they do not alter the governance status of the underlying dataset.

9.6 Controlled Interoperability

The purpose of optional execution environments is not to collapse governance into training logic, but to allow **controlled interoperability** between the two.

Under this model, governance artifacts and dataset passports establish what may be used, under what conditions, and in which lifecycle state. Execution environments then consume those declarations and apply them within training or runtime contexts.

This approach allows organizations to support a variety of deployment patterns, including:

- governance-only onboarding workflows
- internal training pipelines
- Q-Pathformer-compatible capsule workflows
- future architectures not yet defined

As a result, the onboarding framework remains stable even as execution systems evolve.

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10. Governance Boundaries and Non-Claims

The dataset onboarding framework described in this document provides a structured mechanism for transforming governed artifacts into machine-readable datasets while preserving provenance, authorization scope, and lifecycle controls. However, the presence of datasets within this framework does not imply certification, validation, or approval of those datasets.

This section clarifies the governance boundaries of the framework and identifies claims that must **not** be inferred from dataset onboarding or lifecycle routing.

The framework is designed to support traceability, containment, and controlled dataset evolution. It does not replace governance authorities, regulatory oversight, or organizational decision-making processes.

10.1 No Dataset Certification

Dataset passports and lineage records do not certify dataset quality, completeness, or correctness.

Recording a dataset within the onboarding framework does not imply that the dataset has been verified as accurate, unbiased, or suitable for model training. Instead, the passport documents the dataset's provenance and authorization context so that governance authorities may evaluate the dataset appropriately.

Organizations using the framework remain responsible for determining whether datasets meet their own quality or validation standards.

10.2 No Trust or Assurance Claims

The framework does not assign trust levels or assurance classifications to datasets or models.

While lineage records and provenance anchors may help organizations trace the origin of datasets, these records should not be interpreted as indicators of dataset reliability or model safety.

Trust and assurance determinations require independent evaluation processes that fall outside the scope of this document.

10.3 No Compliance Determination

Dataset onboarding does not constitute regulatory compliance or legal approval.

Datasets may originate from environments subject to regulatory or organizational constraints. Recording those datasets within the lifecycle model does not imply that the datasets satisfy any specific regulatory requirement or legal obligation.

Compliance assessments must be conducted through appropriate governance or regulatory procedures.

10.4 No Transfer of Governance Authority

Execution environments consuming datasets from this framework do not acquire governance authority.

Training systems, runtime pipelines, or execution environments may interpret dataset passports and apply lifecycle routing rules, but they do not determine dataset authorization independently. Governance authority remains external to execution systems and must be exercised through organizational policy processes.

This separation preserves the governance structure established in earlier Access-PoD publications.

10.5 No Automatic Dataset Promotion

Datasets do not become durable training datasets automatically.

Lifecycle state transitions, particularly promotion from provisional datasets to durable training corpora, require explicit governance authorization and the issuance of a new dataset passport version.

This requirement prevents datasets from influencing model behavior without appropriate oversight.

10.6 No Implied Model Safety

The presence of governed datasets within a training pipeline does not imply that resulting models are safe, reliable, or appropriate for deployment.

Model safety depends on numerous factors including model architecture, training procedures, evaluation methods, and operational context. The dataset onboarding framework addresses only the governance and lifecycle aspects of datasets used within such pipelines.

10.7 Preservation of Governance Independence

The framework intentionally separates governance artifacts from execution mechanisms.

Execution environments may provide training capabilities, clustering mechanisms, or cryptographic provenance encoding. However, these mechanisms do not alter the governance status of the datasets they process.

Maintaining this separation ensures that dataset governance remains transparent, auditable, and independent of any particular AI implementation.

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11. Implementation Guidance

The dataset onboarding framework described in this document establishes governance structures for dataset provenance, lifecycle management, and authorization routing. Practical implementations of the framework may use software tools to automate the transformation of governance artifacts into machine-readable datasets while preserving the governance controls defined earlier.

This section provides general guidance for implementing the onboarding framework. It does not prescribe specific software architectures or programming languages. Instead, it describes the logical processes that an implementation may follow.

Detailed tooling, scripts, and automation examples may be provided separately within associated repositories.

11.1 Artifact Ingestion

Implementation begins with the ingestion of governance artifacts or other structured records that serve as dataset sources.

Examples of artifact sources include:

- inspection records produced under governance workflows
- simulation outputs or analytical reports
- operational documentation and scenario analyses
- structured research datasets

Artifacts should be preserved in their canonical form before transformation. If artifacts originate from inspection workflows such as ID_3 freeze sessions, the implementation should retain references to the frozen records and their associated hash manifests.

Preserving canonical artifacts ensures that dataset transformations remain traceable to their original sources.

11.2 Dataset Conversion

Once artifacts are ingested, the implementation may perform a dataset conversion step.

During conversion, structured signals contained within artifacts are extracted and organized into machine-readable dataset structures. These signals may include scenario descriptions, analytical observations, governance annotations, or other structured information relevant to model training.

The conversion process should preserve provenance anchors by linking dataset records to their originating artifacts. This linkage allows dataset lineage to remain visible even after transformation into training-ready structures.

Conversion tools may produce provisional datasets that enter the lifecycle pipeline as **State 3 datasets**.

11.3 Dataset Passport Generation

Following conversion, each dataset unit should be assigned a Dataset Passport.

The passport records:

- dataset identity
- provenance anchors
- lifecycle state
- authorization scope
- lineage references

Passport generation provides the governance context required for routing decisions within the lifecycle model.

Implementations may automate passport creation during dataset conversion so that each dataset unit is immediately associated with its governance metadata.

11.4 Lifecycle Routing

After passport generation, datasets are routed according to the lifecycle rules defined earlier in this document.

In most cases, newly created datasets will be routed to **State 3**, where they remain under containment while supporting runtime reasoning, simulation, or synthetic dataset generation.

Datasets may later move into **State 2** during validation or governance review processes. Routing decisions may be informed by attributes recorded in the Dataset Passport, including authorization scope and lineage relationships.

Lifecycle routing mechanisms ensure that provisional datasets do not automatically become durable training corpora.

11.5 Authorization Filtering

Before datasets become eligible for durable training use, implementations should perform authorization filtering.

Authorization filtering ensures that dataset elements exceeding their permitted usage scope are removed or restricted. This process helps enforce governance policies and prevents unauthorized signals from entering persistent training datasets.

Filtering decisions should be guided by the authorization scope recorded in the Dataset Passport. If dataset elements conflict with governance constraints, those elements must be excluded from the dataset version considered for promotion.

Authorization filtering therefore acts as the final containment mechanism before datasets move toward durable training use.

11.6 Dataset Promotion Workflow

When datasets satisfy governance validation requirements, they may be promoted into durable training datasets.

Promotion involves issuing a new dataset version and updating the Dataset Passport to reflect the lifecycle transition. The promotion event should also be recorded within the dataset lineage framework.

Promotion workflows typically involve the following steps:

1. validating dataset provenance and lineage anchors
2. confirming authorization scope
3. applying authorization filtering

4. issuing a new dataset version
5. updating the dataset passport and lineage records

These steps ensure that durable training datasets remain traceable and governance-aligned.

11.7 Optional Execution Integration

Organizations may integrate lifecycle-routed datasets into execution environments capable of performing model training, runtime reasoning, or synthetic dataset generation.

If such environments are used, the Dataset Passport may include optional execution metadata fields that assist the environment in organizing dataset structures or training pipelines.

These metadata fields do not alter governance authority or lifecycle status. They merely provide contextual hints to execution environments about how datasets may be organized or consumed during training processes.

11.8 Repository Tooling

Software implementations supporting this framework may include automation tools such as:

- artifact ingestion utilities
- dataset conversion tools
- dataset passport generators
- lifecycle routing controllers
- authorization filtering mechanisms
- dataset promotion workflows

These tools may be implemented as scripts, services, or modular pipelines depending on the operational needs of the organization.

Detailed examples of such tools may be provided within associated repositories, including implementations compatible with environments such as Q-Pathformer.

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12. Example Dataset Passport

To illustrate how the dataset onboarding framework operates in practice, this section provides an example Dataset Passport derived from an **Access-PoD ID_3 freeze session**.

The example demonstrates how governance artifacts produced during operational reasoning capture can be converted into machine-readable dataset records while preserving provenance anchors, lifecycle state, and authorization scope.

The example dataset originates from a governance stress-testing session examining how reporting of AI-assisted content percentages may shift under operational pressure. The session generated structured reasoning records capturing scenario seeds, stress conditions, fragility points, uncertainty notes, and escalation triggers. These records were later packaged under Access-Mode ID_3 and frozen for inspection-grade archival integrity.

The freeze package includes four canonical artifacts:

- `session_baseline.canonical.json`
- `session_variant.canonical.json`
- `module_03_baseline.canonical.json`
- `module_03_variant.canonical.json`

These artifacts were stamped prior to hashing and archived as part of the ID_3 freeze process. The freeze record also documents the timestamp of the archive event and the folder location of the freeze package.

ID3_FREEZE_NOTE

A hash manifest accompanies the freeze package to preserve artifact integrity. The canonical artifact hashes recorded in the manifest include:

- module baseline record
- module variant record
- session baseline record
- session variant record.

  hashes

These hashes allow auditors or governance authorities to verify that the dataset signals originate from an unchanged inspection artifact.

12.1 Example Dataset Passport Record

The following example illustrates how a Dataset Passport may be generated from the ID_3 freeze package.

```
{
  "dataset_passport_id": "qpds-2026-02-10-m03-001",
```

```
  "dataset_title": "AI Usage Percentage Stress Scenario Signals",
  "dataset_origin": "Access-PoD ID_3 Freeze Session",

  "provenance": {
    "source_mode": "ID_3",
    "freeze_timestamp": "2026-02-28T21:57:07Z",
    "canonical_artifacts": [
      "session_baseline.canonical.json",
      "session_variant.canonical.json",
      "module_03_baseline.canonical.json",
      "module_03_variant.canonical.json"
    ],
    "artifact_hashes": {
      "module_baseline":
"bd543483467ba8091f05b91ffafb96cc72c5a69c26995c211edcdb6cf3e1637c",
      "module_variant":
"7000b9888893866ad781751346293cdf977ed9bab23e46ee6fe8afd958820d41",
      "session_baseline":
"4089536a4c2bd91f822a88d6a9561322db00f333834257919b30fa7ea1036698",
      "session_variant":
"7bf3826e371c91dc0249e6d099a6771033ddc2bd65a09791fc36584c987f9c29"
    }
  },

  "lifecycle_state": "STATE_3",

  "authorization_scope": {
    "runtime_reasoning": true,
    "synthetic_generation": true,
    "fine_tuning": false,
    "external_training": false
  },

  "lineage": {
    "parent_session": "ID2-20260210-1835-session",
    "variant_session": "ID2-20260210-1907-session-timepressure",
    "policy_reference": "APOD-TR-004 Version C"
  },

  "dataset_signals": {
    "signal_type": "governance_reasoning_capture",
    "source_module": "Module 03 - SAPP Simulation & Stress Testing",
    "teacher_signals": [
      "scenario_seeds",
      "stress_conditions",
      "fragility_points",
      "authority_drift_flags"
    ],
    "student_interpretation": "pattern extraction for governance reasoning
signals"
  }
}
```


This passport indicates that the dataset originates from an inspection-packaged artifact but remains in **State 3**, where it may support runtime reasoning or synthetic dataset generation.

Because the artifact records capture operational reasoning signals rather than validated training data, the passport explicitly restricts the dataset from being used for fine-tuning or external training.

12.2 Signal Extraction Example

The dataset signals recorded in the passport originate from the module records contained within the freeze package.

These records capture structured reasoning signals observed during governance stress simulations. Examples of captured signals include:

- scenario seeds describing operational conditions
- stressors such as time pressure or conflicting telemetry
- fragility points identifying governance boundary drift
- uncertainty notes documenting interpretive limits
- escalation triggers identifying governance intervention points.

These signals do not represent governance decisions. Instead, they document reasoning conditions and interpretive signals that may later be used for exploratory learning, runtime reasoning, or synthetic dataset generation.

The passport therefore preserves the contextual meaning of the signals while preventing their automatic promotion into durable training datasets.

12.3 Lifecycle Interpretation

Within the lifecycle model defined earlier in this document, the example dataset is classified as a **State 3 dataset**.

State 3 datasets represent provisional dataset resources that may support runtime reasoning or synthetic dataset preparation while remaining under governance containment.

Promotion to **State 1 durable training datasets** would require:

- governance validation
- authorization scope update
- dataset filtering and transformation
- issuance of a new dataset passport version.

This requirement ensures that exploratory reasoning signals captured in governance artifacts do not automatically influence persistent model behavior.

12.4 Optional Execution Metadata

If the dataset were to be consumed by an execution environment such as Q-Pathformer, the Dataset Passport could optionally include execution metadata describing clustering or capsule organization.

Such metadata might include:

- dataset grouping identifiers
- cluster classification tags
- training stage indicators
- capsule identifiers

These fields assist execution environments in organizing datasets for training but do not alter the governance semantics of the passport.

Execution metadata therefore remains optional and may be omitted when datasets are used in governance-only workflows.

12.5 Traceability Outcome

The example demonstrates how the onboarding framework preserves traceability between:

- governance artifacts
- frozen inspection records
- dataset signals
- machine-readable dataset passports.

This traceability ensures that datasets derived from governance reasoning artifacts remain interpretable and auditable even after transformation into structured training resources.

---

13. Future Work

The dataset onboarding framework described in this document establishes a governance-aligned mechanism for transforming artifacts into machine-readable datasets while preserving

provenance, lifecycle state, and authorization boundaries. Future development of the framework may explore several extensions that strengthen the relationship between dataset governance, model training, and runtime learning systems.

13.1 Dataset Converter and Authorization Tooling

Future implementations may provide automated tools to support the processes described in Section 11.

Such tools may include:

- artifact ingestion utilities for converting governance artifacts into dataset records
- dataset converters that extract signals from inspection artifacts
- dataset passport generators that create machine-readable governance records
- lifecycle routing controllers that assign datasets to State 1, State 2, or State 3
- authorization filtering mechanisms that remove signals exceeding permitted usage scope.

These tools may be implemented within associated repositories to support reproducible dataset onboarding workflows.

13.2 Reference Implementations for Execution Environments

While the onboarding framework is execution-agnostic, future work may include reference implementations demonstrating how governed datasets can be integrated into AI training systems.

One potential reference environment is **Q-Pathformer**, which supports modular dataset organization, capsule-based training pipelines, and structured dataset provenance.

Future work may explore how dataset passports can be interpreted by execution environments to support:

- dataset clustering and grouping
- staged training preparation
- synthetic dataset generation
- runtime reasoning support.

These implementations would remain optional and would not alter the governance semantics of the onboarding framework.

13.3 Three-State Dataset Learning Model

Future research may further explore the relationship between the dataset lifecycle model and machine learning architectures.

Future implementations may also explore execution environments inspired by quantum or quantum-inspired machine learning architectures, including multi-state learning systems capable of organizing datasets across exploratory, validation, and durable training stages.

One potential direction is a three-state dataset learning model aligned with the lifecycle states defined earlier in this document and conceptually related to the multi-state machine learning architecture described in Q-Pathformer research.

In such a model:

- **State 3 datasets** function as contextual memory and exploratory signal sources during runtime sessions.
  These datasets may capture reasoning signals, scenario patterns, or contextual observations that support adaptive learning and synthetic dataset generation.
- **State 2 datasets** serve as a convergence layer where teacher and student signals are evaluated, validated, and organized before durable learning occurs.
- **State 1 datasets** represent curated training corpora used for refined fine-tuning of models under explicit governance authorization.

Under this approach, datasets derived from governance artifacts could be used dynamically during operational sessions while remaining traceable to their original provenance anchors.

13.4 Client and Sector-Specific Exemplars

Future work may also explore **user or client exemplar models** derived from governance artifacts and operational sessions.

Such exemplars may represent domain-specific governance reasoning patterns within particular industries or regulatory environments.

For example, an organization operating in a specific sector may maintain curated State 1 datasets representing validated governance reasoning patterns relevant to that sector. During runtime sessions, State 3 datasets could provide contextual memory signals reflecting recent operational conditions or evolving regulatory interpretations.

This layered structure could allow organizations to refine domain-specific governance models while maintaining reproducible provenance and dataset lineage.

13.5 Context Window Optimization Across Dataset States

Future work may also examine how dataset lifecycle states interact with model context windows.

One potential approach is to maintain shorter context windows within individual lifecycle states while enabling larger contextual understanding across sectors or industries through aggregated datasets.

In this model:

- State-specific datasets provide focused signals within bounded contexts.
- Aggregated sector datasets provide broader contextual awareness across industry domains.

Such approaches may help balance computational efficiency with contextual depth in governance-aware AI systems.

13.6 Continuous Dataset Evolution

Finally, future work may explore methods for continuously updating governed datasets as regulatory environments, operational practices, and sector expectations evolve.

These methods may include:

- periodic dataset promotion reviews
- sector-specific dataset updates
- governance-driven dataset revisions triggered by policy changes
- controlled retraining of models using updated State 1 datasets.

By preserving dataset lineage and lifecycle traceability, such updates can occur without losing historical evidence of how models were trained or how governance practices evolved over time.

13.7 Reference Implementations and Release Tiers

Future releases associated with the Access-PoD research program may include additional implementation artifacts and reference environments supporting the governed dataset lifecycle model.

These releases may include research and educational reference implementations, strategic governance tooling, SPDX-aligned operational models, and multimodal training environments supporting modular agent architectures.

Examples under exploration include multimodal agent frameworks with configurations such as 16-agent, 64-agent, and 144-agent architectures, as well as associated training code and model weight artifacts distributed through separate repositories or releases.

These implementations remain optional and do not alter the governance semantics defined in this document.

Closing Perspective

The framework presented in this document represents the next step in the evolution of the Access-PoD research series. **Version A** established the conceptual foundation for engineering trust through runtime evidence, simulation-driven governance, and telemetry-derived assurance. **Version B** introduced an optional architecture illustration demonstrating how governance controls may interact with execution environments without altering the model-agnostic nature of the framework. **Version C** extended this foundation by defining an onboarding and operational transition discipline designed to manage uncertainty as AI systems move from experimental contexts into controlled runtime governance.

This report builds on those foundations by introducing a structured mechanism for transforming governance artifacts into machine-readable datasets through the **Dataset Passport and lifecycle routing model**. Together, these publications describe a coherent governance progression: from conceptual trust engineering, to architecture-aware governance, to operational onboarding discipline, and finally to governed dataset lifecycle management. Each layer remains compatible with diverse AI architectures while reinforcing the central principle that trustworthy AI systems must be supported by **continuous evidence, traceable artifacts, and accountable operational controls** rather than static documentation alone.

References

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Vol.1 — Q-Pathformer: Multi-state Machine Learning Architecture
QPATH-ARCH-001 - *Q-Pathformer - Multi-state Machine Learning and LLM Training.pdf*

Policy of Developments – Access PoD - series

Vol.1 - A WEB of Sharing Trust-responsibility toward a trustless future.

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Vol.2B - TR-003 — Architecture Illustration
Vol.2C - TR-004 — Operational Onboarding Discipline
Vol.2D - TR-005 — Governance Stress Testing (planned)
Vol.2E - TR -006 — Execution Governance Extensions (planned)
TR-007 — Governed Dataset Lifecycle Framework